

Exp no: 13 Determination and plotting of intrinsic impedance and phase velocity of various medium

Date:

Aim:

To determine and plot intrinsic impedance and phase velocity of various medium.

Software required:

- SCILAB version 6.0.1

Theory:

Intrinsic Impedance (Z)

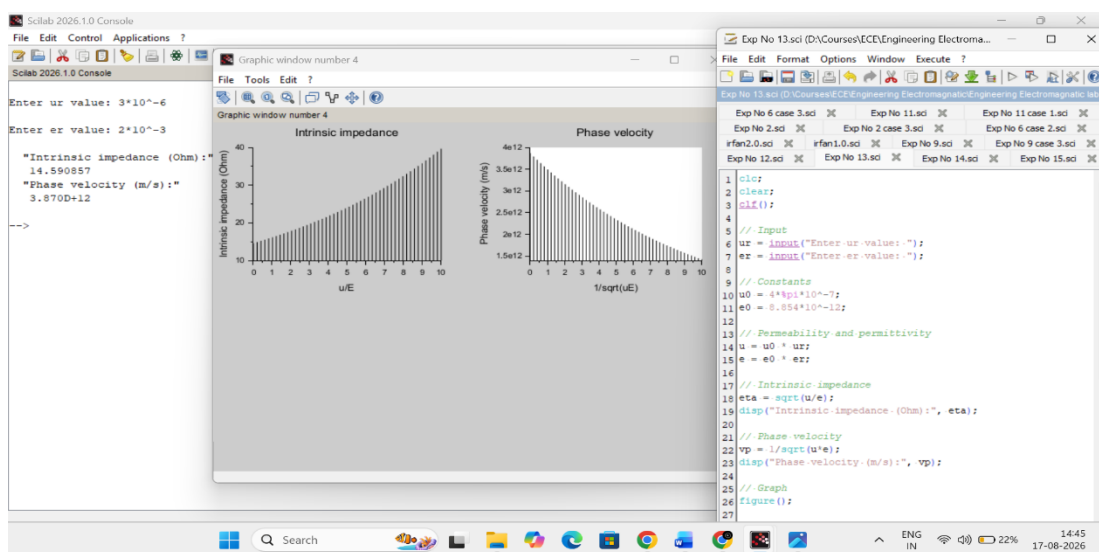
The **intrinsic impedance** of a medium is an important parameter that characterizes the relationship between the electric and magnetic fields in an electromagnetic wave propagating through that medium. It represents the opposition that the medium presents to the electric and magnetic fields.

- **Definition:** Intrinsic impedance Z is the ratio of the electric field (E) to the magnetic field (H) in an electromagnetic wave propagating in that medium. In other words, it describes how much resistance the wave experiences as it propagates.

The intrinsic impedance of a medium can be expressed mathematically as:

$$Z = \sqrt{\frac{\mu}{\epsilon}} \quad \text{Where:}$$

- Z is the intrinsic impedance of the medium in ohms (Ω),
- μ is the permeability of the medium in henries per meter (H/m),
- ϵ is the permittivity of the medium in farads per meter (F/m)



Exr N^o. 13

Determination of Intrinsic Impedance and
phase velocity

$$\mu_r = 3 \times 10^{-6}$$

or

$$\epsilon_r = 2 \times 10^{-3}$$

Intrinsic Impedance

$$\eta = \sqrt{\frac{\mu}{\epsilon}}$$

where

$$\mu = \mu_0 \mu_r$$

$$\epsilon = \epsilon_0 \epsilon_r$$

$$\eta = \sqrt{\frac{\mu_0 \mu_r}{\epsilon_0 \epsilon_r}}$$

$$= \sqrt{\frac{(4\pi \times 2.14 \times 10^{-7}) (3 \times 10^{-6})}{8.854 \times 10^{-12}}}$$

$$= \sqrt{\frac{3.768 \times 10^{-12}}{8.854 \times 10^{-12}}}$$

$$\eta = 176.62 \Omega$$

$$\eta = \sqrt{21289}$$

$$\boxed{\eta = 14.59}$$

Phase velocity

$$V_p = \frac{1}{\sqrt{\mu \epsilon}}$$

$$= \frac{1}{\sqrt{(\mu_0 \mu_r)(\epsilon_0 \epsilon_r)}}$$

$$= \frac{1}{\sqrt{(3.796 \times 10^{-12})(1.7708 \times 10^{-4})}}$$

$$V_p = 3.87 \times 10^8 \text{ m/s}$$

$$\boxed{V_p = 3.87 \times 10^8 \text{ m/s}}$$

Result:

Thus, the determination and plotting of intrinsic and phase velocity is performed and successfully verified.

Case 1: To determine the intrinsic impedance and phase velocity of electromagnetic waves in free space (vacuum).

Analytical:

Calculate the Intrinsic Impedance and Phase Velocity of Electromagnetic Waves in Free Space (Vacuum)

Given:

- **Material:** Free space (vacuum)
- **Permeability of free space** ($\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$)
- **Permittivity of free space** ($\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$)

Solution:

The **intrinsic impedance** Z_0 of free space (vacuum) is given by the formula:

$$Z_0 = \sqrt{\frac{\mu_0}{\epsilon_0}}$$

Where:

- $\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$ (permeability of free space),
- $\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$ (permittivity of free space).

Substitute the known values of μ_0 and ϵ_0 into the formula:

$$Z_0 = \sqrt{\frac{4\pi \times 10^{-7}}{8.854 \times 10^{-12}}}$$

First, calculate the ratio inside the square root:

$$\frac{4\pi \times 10^{-7}}{8.854 \times 10^{-12}} \approx 1.132 \times 10^5$$

Now, take the square root:

$$Z_0 \approx \sqrt{1.132 \times 10^5} \approx 377 \Omega$$

So, the **intrinsic impedance** of free space is approximately:

$$Z_0 \approx 377 \Omega$$

The phase velocity v_p of electromagnetic waves in free space is given by:

$$v_p = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

Substitute the known values of μ_0 and ϵ_0 :

$$v_p = \frac{1}{\sqrt{(4\pi \times 10^{-7})(8.854 \times 10^{-12})}}$$

First, calculate the product inside the square root:

$$(4\pi \times 10^{-7}) \times (8.854 \times 10^{-12}) \approx 1.112 \times 10^{-18}$$

Now, take the square root:

$$\sqrt{1.112 \times 10^{-18}} \approx 1.055 \times 10^{-9}$$

Thus, the phase velocity is:

$$v_p = \frac{1}{1.055 \times 10^{-9}} \approx 3 \times 10^8 \text{ m/s}$$

So, the phase velocity of electromagnetic waves in free space is approximately:

$$v_p = 3 \times 10^8 \text{ m/s}$$

This is the speed of light in a vacuum, which is consistent with the known value for c .

The screenshot shows the Scilab 2026.1.0 interface. The console window on the left displays the output of the script: "Intrinsic Impedance (eta0) in ohms: 376.73431" and "Phase Velocity (vp) in m/s: 2.998D+08". The editor window on the right shows the following code:

```

1 clc;
2 clear;
3
4 // Constants
5 mu0 = 4 * pi * 10^(-7); // Permeability of free space (H/m)
6 epsilon0 = 8.854 * 10^(-12); // Permittivity of free space (F/m)
7
8 // Calculate intrinsic impedance
9 eta0 = sqrt(mu0 / epsilon0);
10
11 disp("Intrinsic Impedance (eta0) in ohms: " + string(eta0));
12
13 // Calculate phase velocity
14 vp = 1 / sqrt(mu0 * epsilon0);
15
16 disp("Phase Velocity (vp) in m/s: " + string(vp));
17

```

Result:

Thus the determination and plotting of intrinsic and phase velocity is performed and successfully verified in free space.

Case 2: To determine the intrinsic impedance and phase velocity for a wave propagating through a conductive medium like copper.

Analytical: Calculate the Intrinsic Impedance and Phase Velocity of Electromagnetic Waves in Copper

Given:

- **Material:** Copper (conductive medium)
- **Relative Permittivity (ϵ_r)** of copper: 1 (approximately, since copper is a good conductor and its permittivity is close to the permittivity of free space),
- **Relative Permeability (μ_r)** of copper: 1 (approximately, as copper is a non-magnetic material),
- **Conductivity (σ)** of copper: $5.8 \times 10^7 \text{ S/m}$,
- **Frequency of the wave (f):** $1 \text{ GHz} = 10^9 \text{ Hz}$,
- **Permeability of free space (μ_0):** $4\pi \times 10^{-7} \text{ H/m}$,
- **Permittivity of free space (ϵ_0):** $8.854 \times 10^{-12} \text{ F/m}$.
- $\mu_r = 1$,
- $\epsilon_r = 1$,
- $\sigma = 5.8 \times 10^7 \text{ S/m}$,
- $\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$,
- $\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$,
- $\omega = 6.2832 \times 10^9 \text{ rad/s}$.

We first calculate μ and ϵ :

$$\mu = \mu_0 \mu_r = 4\pi \times 10^{-7} \times 1 = 1.2566 \times 10^{-6} \text{ H/m}$$

$$\epsilon = \epsilon_0 \epsilon_r = 8.854 \times 10^{-12} \times 1 = 8.854 \times 10^{-12} \text{ F/m}$$

Now, calculate the term $\frac{\sigma}{\omega\epsilon}$:

$$\frac{\sigma}{\omega\epsilon} = \frac{5.8 \times 10^7}{6.2832 \times 10^9 \times 8.854 \times 10^{-12}} \approx 1.039 \times 10^3$$

The complex intrinsic impedance becomes:

$$Z = \sqrt{\frac{1.2566 \times 10^{-6}}{8.854 \times 10^{-12}}} (1 + j \times 1.039 \times 10^3)$$

First, calculate the magnitude of the impedance without the complex term:

$$\sqrt{\frac{1.2566 \times 10^{-6}}{8.854 \times 10^{-12}}} = \sqrt{1.419 \times 10^5} \approx 377 \Omega$$

Thus, the complex intrinsic impedance Z is:

$$Z \approx 377 (1 + j \times 1.039 \times 10^3)$$

The phase velocity v_p of the wave in a conducting medium is given by the equation:

$$v_p = \frac{1}{\sqrt{\mu\epsilon}} \cdot \frac{1}{\sqrt{1 + \frac{\sigma^2}{\omega^2\epsilon^2}}}$$

However, since copper is a good conductor, the phase velocity will be reduced significantly due to the skin effect. To simplify the calculation for good conductors, the phase velocity can be approximated by:

$$v_p \approx \frac{1}{\sqrt{\mu\epsilon}}$$

Thus, using the values for μ and ϵ calculated earlier:

$$v_p = \frac{1}{\sqrt{1.2566 \times 10^{-6} \times 8.854 \times 10^{-12}}}$$

$$v_p = \frac{1}{\sqrt{1.112 \times 10^{-17}}} \approx 3 \times 10^8 \text{ m/s}$$

The screenshot shows two windows from the Scilab 2026.1.0 environment. The left window is the 'Console', which displays the results of several commands: the intrinsic impedance η is calculated as $0.01428998 + j \cdot 0.0142898$ Ohms, its magnitude is 0.0202088 Ohms, its phase is 45 degrees, and the phase velocity (using a lossless approximation) is 2.998×10^8 m/s. The right window is the 'Editor', showing a script for 'Exp No 13 case 2.sci'. The script defines the imaginary unit j , calculates the intrinsic impedance η of a conducting medium using the formula $\eta = \sqrt{j \omega \mu / (\sigma + j \omega \epsilon)}$, and then calculates its magnitude and phase. It also calculates the phase velocity v_p using the lossless approximation $v_p = 1 / \sqrt{\mu \epsilon}$. The script includes comments in green and uses `disp` to show the results at various steps.

Result:

Thus the determination and plotting of intrinsic and phase velocity is performed and successfully verified in copper.

Case 3: To determine and compare the intrinsic impedance and phase velocity of electromagnetic waves propagating through different media: Vacuum, Air, and Water.

Analytical:

Determine and compare the intrinsic impedance and phase velocity of electromagnetic waves propagating through different media

Conditions:

- **Vacuum:**
 - Permeability: $\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$
 - Permittivity: $\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$
- **Air (Assumed to be similar to Vacuum):**
 - Permeability: $\mu_{\text{air}} \approx \mu_0$
 - Permittivity: $\epsilon_{\text{air}} \approx \epsilon_0$
- **Water (Dielectric Material):**
 - Permeability: $\mu_{\text{water}} \approx \mu_0$
 - Relative Permittivity: $\epsilon_{\text{water}} = 80 \times \epsilon_0$

Solution:

1. Phase Velocity for Vacuum:

For vacuum ($\mu_r = 1$ and $\epsilon_r = 1$):

$$v_{p_{\text{vacuum}}} = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = \frac{1}{\sqrt{(4\pi \times 10^{-7})(8.854 \times 10^{-12})}}$$

First, calculate the product inside the square root:

$$(4\pi \times 10^{-7}) \times (8.854 \times 10^{-12}) \approx 1.112 \times 10^{-17}$$

Now, calculate the phase velocity:

$$v_{p_{\text{vacuum}}} = \frac{1}{\sqrt{1.112 \times 10^{-17}}} \approx 3 \times 10^8 \text{ m/s}$$

Thus, the phase velocity in vacuum is:

$$v_{p_{\text{vacuum}}} \approx 3 \times 10^8 \text{ m/s}$$

2. Phase Velocity for Air:

For air, the phase velocity is nearly identical to that in vacuum since air has similar properties to vacuum ($\mu_r = 1$ and $\epsilon_r = 1$).

Thus, the phase velocity in air is approximately:

$$v_{p_{\text{air}}} \approx 3 \times 10^8 \text{ m/s}$$

3. Phase Velocity for Water:

For water ($\mu_r = 1$ and $\epsilon_r = 80$):

$$v_{p_{\text{water}}} = \frac{1}{\sqrt{\mu_0 \epsilon_{\text{water}}}} = \frac{1}{\sqrt{(4\pi \times 10^{-7})(7.0832 \times 10^{-10})}}$$

First, calculate the product inside the square root:

$$(4\pi \times 10^{-7}) \times (7.0832 \times 10^{-10}) \approx 8.912 \times 10^{-16}$$

Now, calculate the phase velocity:

$$v_{p_{\text{water}}} = \frac{1}{\sqrt{8.912 \times 10^{-16}}} \approx 1.06 \times 10^8 \text{ m/s}$$

Thus, the phase velocity in water is:

$$v_{p_{\text{water}}} \approx 1.06 \times 10^8 \text{ m/s}$$

The screenshot shows the Scilab 2026.1.0 environment. The console on the left displays the results of calculations for Vacuum, Air, and Water. The editor on the right shows the corresponding Scilab script.

Console Output:

```

"Vacuum:"
"Intrinsic Impedance (eta) = 376.81944 Ohm"
"Phase Velocity (vp) = 2.999D+08 m/s"
"Air:"
"Intrinsic Impedance (eta) = 376.81944 Ohm"
"Phase Velocity (vp) = 2.999D+08 m/s"
"Water:"
"Intrinsic Impedance (eta) = 42.129694 Ohm"
"Phase Velocity (vp) = 33525745 m/s"
-->

```

Editor Code (Exp No 13 case 3.sci):

```

30 disp("Intrinsic Impedance (eta) = " + string(eta_vacuum) +
31 " Ohm");
32 disp("Phase Velocity (vp) = " + string(vp_vacuum) + " m/s");
33 // Calculate for Air
34 [eta_air, vp_air] = calculate_properties(mu_air, epsilon_a
35 ir);
36 disp("Air:");
37 disp("Intrinsic Impedance (eta) = " + string(eta_air) + ".
38 Ohm");
39 disp("Phase Velocity (vp) = " + string(vp_air) + " m/s");
40 // Calculate for Water
41 [eta_water, vp_water] = calculate_properties(mu_water, eps
42 ilon_water);
43 disp("Water:");
44 disp("Intrinsic Impedance (eta) = " + string(eta_water) +
45 " Ohm");
46 disp("Phase Velocity (vp) = " + string(vp_water) + " m/s");

```

Result: Thus, the determination and plotting of intrinsic and phase velocity is performed and successfully verified in copper.